SECOND SEMESTER B.TECH,EXAMINATION, JULY-2023

USN:_____ ENVIRONMENTAL SCIENCES [22CSBS121]

SL.NO	QUESTIONS	A	B	C	D	Correct Answers
1	French word "Environ" which means	Enclosed	Encircle	Outskirt	Network	
2	"Atmo" refers to	steam and vapor	gases	Aerosol	rocks and minerals	
3	Hydrosphere comprises of all the----------	Forest resources	Grassland resources	Mineral and rocks	Water resources	
4	Approximately 2% of total water is in the form of	Vapor	Glaciers	Ground water	Lakes	
5	The earth's surface is composed of ----types of lithospheres	Three	Two	Four	Six	
6	Area that comprises all living species as well as their products is known as-------	Lithosphere	Stratosphere	Mesosphere	Biosphere	
7	Biosphere spans around-------miles from top to bottom	10 miles	12 miles	14 miles	15 miles	
8	----------is defined as structural and functional unit of the biosphere	Biosystem	Ecosystem	Biodiversity	Lithosphere	
9	Abiotic components refers to	Non-Living things	Living things	Both of these	None of these	
10	Organisms which produces its source of energy is known as----	Lithotrophes	Archeatrophs	Autotrophes	Organotrophes	
11	Photoautotrophes uses-----energy to produce food	Solar	Tidal	Hydro	Light	
12	Nitrobacter is an example of	Methanogenic bacteria	Chemoautotrophes	Photoautotrophes	Lithotrophes	

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13	Which of the following is a decomposer	Vibrio cholerae	Staphylococcus aureus	Escherichia coli	Saprophytes	
14	Which of the following is a main function of ecosystem	Regulate and to promote the ecological processes	Interferes with the ecological processes	Controls the ecological processes	Rearrange the ecological processes	
15	Energy is the fundamental factor that drives all	Physical processes	Chemical processes	Metabolic processes	Thermodynamic processes	
16	Energy flow in ecosystem is	Multidirectional	Unidirectional	Continuous	Random	
17	Each nutritional level in the food chain is referred as	Food level	Energy level	Trophic level	Linear level	
18	First law of thermodynamics can be exemplified by	Respiration	Excretion	Photosynthesis	Food pyramid	
19	Energy transformation is not 00% efficient is stated by	Second law of thermodynamics	First law of thermodynamics	Zeroth law of thermodynamics	Law of ecosystem	
20	Which of the following statement is related to food chain	Transfer of food energy from producer through a series of organisms with repeated eating	Transfer of food energy from chemoautotrophs to a series of organisms	Transfer of food energy from abiotic component to a series of organisms	Transfer of food energy from Herbivores to a series of organisms with repeated eating and being eaten	
21	Biodiversity refers to living things and their------	Relationships	Interactions	Overview	Understandings	
22	_____Million species have been identified and described so far by the scientists?	1.2 million	1.8 million	8.7 million	7.8 million	
23	Biodiversity is explored in--------levels	two	three	four	multilevel	
24	Variety of genes within a species is known as -------	Genetic diversity	Species diversity	Ecosystem diversity	None of these	
25	Variety of species within a habitat or region is referred as --	Biodiversity	Species diversity	Eco diversity	Environmental balance	

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26	Ecosystem diversity is the variety of ecosystems in a -----	Given place	Country	Ocean	Forest	
27	Biogeography is the study of the---------- of species	Relationship	Understanding	Communication	Distribution	
28	India's natural diversity------ ranks place in Asia	3rd	4th	5th	10th	
29	Documented floral diversity in India is--------	17500	6200	7500	10500	
30	Isabgol is obtained from husk of----------	Plantago	Cinchona	Marijuna	None of these	
31	Penicillin is a----------	Antiviral	Anti analgesic	Anti pyretic	Antibiotic	
32	Musk, Silk, Wool is the example of----- value of biodiversity	Protective	Productive	Ethical	Aesthetic	
33	Which of the following show minimum diversity?	Dessert	Ocean	Forest	Mountain	
34	India is one of the-------- mega diversity country	11	12	13	14	
35	Rate of change across large composition gradients is known as----------	Beta richness	Alpha richness	Gamma richness	Point richness	
36	Species which are restricted only to a particular area is known as _	Endemic Species	Restricted Species	Particular Species	None of these	
37	Which megadiversity country has 7500 Km long coastline	New Zealand	India	China	Saudi Arabia	
38	50%----------are endemic to India	Tigers	Elephants	Lizards	Snakes	

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39	Which region in India has maximum endemism and diversity?	Gangetic region	Himalayan region	Western Ghats	Aravalli Mountain region	
40	Biogeographic region with a significant reservoir of biodiversity that is under threat from humans is called as----------	Hotspring Biodiversity	Hotspot Biodiversity	Geographic Biodiversity	Reservoir Biodiversity	
41	Biosphere consists of	Lithosphere	Atmosphere	Hydrosphere	All the above	
42	Component of biosphere related with soil is	Lithosphere	Hydrosphere	Atmosphere	None of the above	
43	Source of energy in the biosphere is	Producer	Decomposer	Sunlight	Heat of earth.	
44	Ozone layer occurs in	Ionosphere	Stratosphere	Thermosphere	Troposphere	
45	Which of the following is a nonrenewable energy resource?	Solar	Methane	Hydroelectric	Coal	
46	Which one of the following is a branch of forestry?	Protection Forestry	Deforestation	Forest Management	Restricted Forestry	
47	This group consists of non-renewable organic resources	Water, air and minerals	Natural gas, oil and coal	Wood, water and natural pastures	Sand, air and clay	
48	The major non-renewable energy usage in India is	Coal	Petroleum and other liquids.	Natural gas	Nuclear	
49	Which of the following is a disadvantage of renewable energy?	High pollution	Available only in few places	High running cost	Unreliable supply	
50	The ocean covers..... percentage of earth surface	51%	61%	71%	91%	
51	Atmospheric humidity is measured by	Auxanometer	Photometer	Hygrometer	None.	

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52	Which gas likely to be reduced in the atmosphere by deforestation?	Carbon dioxides	Nitrogens	Oxygen	Sulphur dioxides	
53	Which of the following plants are used for producing biofuels	Castor	Rapeseed	Jatropha	all of these	
54	The type of forests grown in the Himalayan mountain region is called:	Broad-leaved forests.	Coniferous forests	Deciduous forests	None of the above	
55	Rooftop rainwater harvesting is a technique to recharge	Ground water	River water	Lake water	Sea water	
56	Which of the following energy resources is renewable?	Coal.	Hydroelectric power.	Natural gas.	Petroleum	
57	Which of the following is not a fossil fuel?	Coal.	Natural gas.	Petroleum.	Uranium.	
58	What form of energy is radiated by the earth and trapped by greenhouse gasses to cause global warming?	Radio waves.	Infrared light..	Ultraviolet light.	Visible light.	
59	Which one of the following type of resource is iron ore.	Renewable	Biotic	Flow	Non-renewable	
60	The energy produced by the hydel-power plant is	Non-polluting and non-renewable	Polluting and non-renewable	Non-polluting and renewable	Polluting and renewable.	
61	Which of the following is secondary pollutant.	NO2	CO	VOM	H2SO4	
62	PAN is	Peroxyacetyl Nitrate	Primary acid nitrate	Peroxy additive nitrogen	None of the above	
63	Sedimentation in waste water treatment is a part of	Primary treatment	Secondary treatment	Tertiary treatment	Sedimentation is not followed	
64	Noise pollution can cause	Hypertension	Reduced sensitivity	Insomnia	All the above	

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65	Which radiation doesn't enter the skin.	Alpha particle	Beta particle	X-rays	Gamma radiation	
66	Particles can go inside the skin but cannot damage tissues.	Alpha particle	Beta particle	X-rays	Gamma radiation	
67	Greenhouse effect is related to	Air pollution	Water pollution	Noise pollution	Soil Pollution	
68	Sewage mostly contains	Non-biodegradable pollutant	Biodegradable pollutant	Effluents	Air pollutants	
69	The chief source of soil and water pollution is	Mining	Agro-industry	Thermal power plant	All of the above	
70	The organisms that break down dead or organic matter and control pollution are	Bacteria	Fungi	Both (i) & (ii)	None of the above	
71	Ozone is entirely different from O_2 as	Is present all over the atmosphere / or	Is acidic	Is a dead poison	All the above	
72	The gas(es) not responsible for ozone depletion:	NO and freons	SO_2	CO_2	Both (b) and (c)	
73	The region containing water vapor is	Thermosphere	Stratosphere	Troposphere	Mesosphere	
74	The global climate is threatened by the increase in the concentration of	Oxygen	Nitrogen	Water vapors	Greenhouse gases	
75	Which of the following would be natural pollution?	Forest fires	Eutrophication	Occurrence of smog	Occurrence of NO2 in the atmosphere	
76	Which of the following statements is incorrect?	Carbon monoxide is the major environmental pollutant	All pollutants are not waste	Nitrogen and Mg can pollute water	Lichens are affected by sulfur dioxide	
77	The soil pollutants that affect the food chain and food web by killing microorganisms and plants are	Pathogens	Chemical fertilizers	Agricultural waste	Pesticides	

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78	The presence of ozone (O3) in the atmosphere of earth	Hinders higher rate of photosynthesis	Helps in checking the penetration of ultraviolet rays on earth	Has been responsible for increasing the average global temperature in	It is advantageous since it supplies O2 for people traveling on jets.	
79	Chlorofluorocarbon is used in	Refrigerators	Air conditioners	Perfumes	All	
80	Which of the following causes an outbreak of jaundice?	Air pollution	Water pollution	Thermal pollution	Soil pollution	
81	Which of the following gases is primarily responsible for the depletion of the ozone layer?	Carbon dioxide (CO2)	Nitrous oxide (N2O)	Chlorofluorocarbons (CFCs)	Methane (CH4)	
82	What is the main cause of the formation of the "ozone hole" over Antarctica?	Industrial pollution	Volcanic activity	Ultraviolet radiation from the sun	Depletion of stratospheric ozone by human-made chemicals	
83	How does land-use planning contribute to disaster mitigation?	Enhancing community preparedness for future disasters	Providing financial compensation to affected individuals	Assessing the extent of the damage caused by a disaster	Reducing exposure to hazards by implementing zoning regulations and building codes to ensure safe construction practices	
84	What are the main adverse effects of ozone layer depletion?	Increased global warming	Skin cancer and cataracts	Acid rain	Depletion of marine life	
85	What is the primary cause of acid rain?	Industrial pollution	Natural volcanic activity	Excessive use of fertilizers in agriculture	Increased solar radiation	
86	Which gases are primarily responsible for the formation of acid rain?	Carbon dioxide (CO2) and methane (CH4)	Nitrogen oxides (NOx) and sulfur dioxide (SO2)	Ozone (O3) and hydrochlorofluorocarbons (HCFCs)	Methane (CH4) and volatile organic compounds (VOCs)	
87	How does acid rain affect the environment?	Enhances the growth of plants and crops	Increases soil fertility and nutrient availability	Harms aquatic ecosystems and damages forests	Improves air quality and reduces pollution levels	
88	What are the main sources of sulfur dioxide (SO2) and nitrogen oxides (NOx) that contribute to acid rain?	Industrial emissions and combustion of fossil fuels	Volcanic eruptions and geothermal activity	Natural forest fires and organic decomposition	Agricultural activities and livestock farming	
89	When did the Chernobyl nuclear disaster occur?	1986	1979	1991	2000	
90	Which country was the location of the Chernobyl nuclear power plant?	Ukraine	Russia	Belarus	Lithuania	

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91	What was the cause of the Chernobyl nuclear disaster?	Earthquake	Terrorist attack	Equipment malfunction during a safety test	Natural disaster	
92	When did the Fukushima nuclear disaster occur?	2005	2011	2015	2020	
93	Which country was the location of the Fukushima nuclear power plant?	Japan	China	South Korea	Taiwan	
94	What was the cause of the Fukushima nuclear disaster?	Earthquake and tsunami	Flood	Terrorist attack	Nuclear reactor malfunction	
95	Which of the following is not one of the three pillars of sustainable development?	Economic development	Social progress	Environmental preservation	Technological advancement	
96	What does the term "sustainable" mean in the context of sustainable development?	Meeting the needs of the present generation without compromising future generations	Utilizing resources at a rapid pace to promote economic growth	Prioritizing short-term gains over long-term environmental impacts	Focusing solely on environmental conservation	
97	Which of the following is not an example of sustainable development?	Usage of wind energy	Usage of solar energy	Crop rotation	Burning fossil fuels	
98	What is the current estimated global population as of 2023?	2 billion	5 billion	8 billion	10 billion	
99	Which among the following countries has the highest population?	India	USA	Australia	Brazil	
100	What factors contribute to population growth of a country?	Birth rate and immigration	Birth rate and emigration	Death rate and immigration	Death rate and emigration	